PORTFOLIO 1 · DATA ANALYTICS

Executive Sales Dashboard

Performance review of Superstore retail sales, 2015–2018

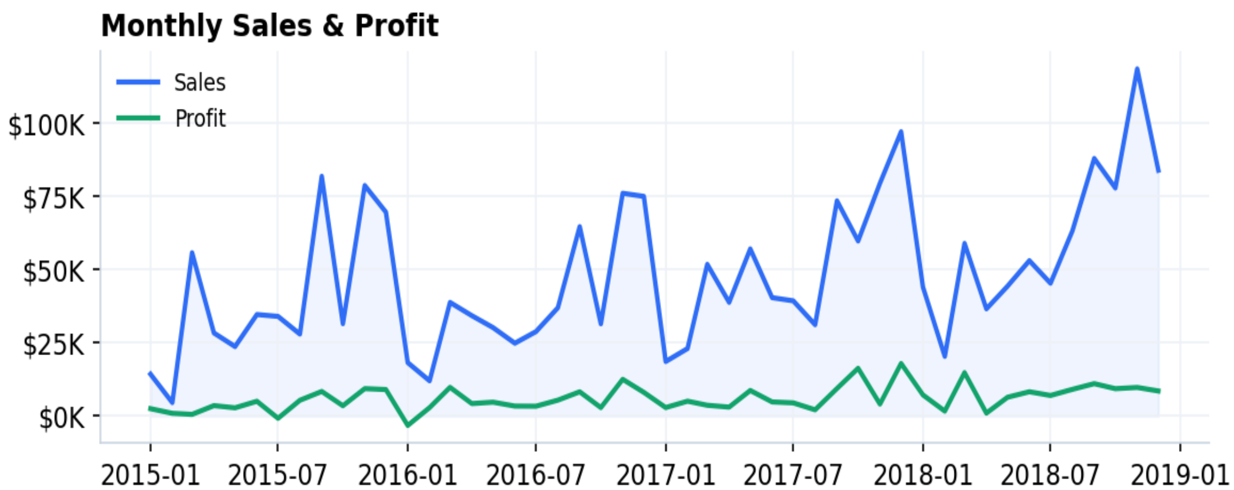
$2.30M	$286K	5,009	12.5%	$459
REVENUE	PROFIT	ORDERS	PROFIT MARGIN	AVG ORDER

Project Overview

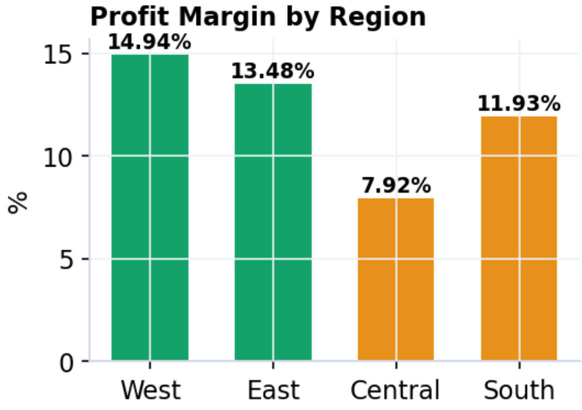
This dashboard gives the CEO and Sales Director a single view of revenue, profit, and order performance across regions, categories, customers, and products. The analysis is built on the Superstore retail dataset covering four years of US orders (2015–2018) across three categories: Furniture, Office Supplies, and Technology.

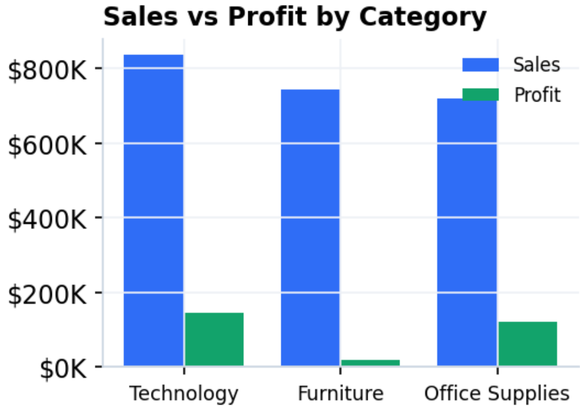
Method. Raw data of 10,800 rows was cleaned to 9,994 valid records by removing 504 duplicate rows and 302 invalid junk rows, standardising dates and currency formats, and normalising product names. Cleaned data was analysed in Python (pandas) and modelled in SQL (PostgreSQL), then surfaced in an interactive dashboard.

Performance at a Glance



Revenue grows year over year with a clear seasonal peak every Q4 (Nov–Dec). Profit tracks sales but stays thin — the gap between the two lines is the core issue this report addresses.





Key Business Insights

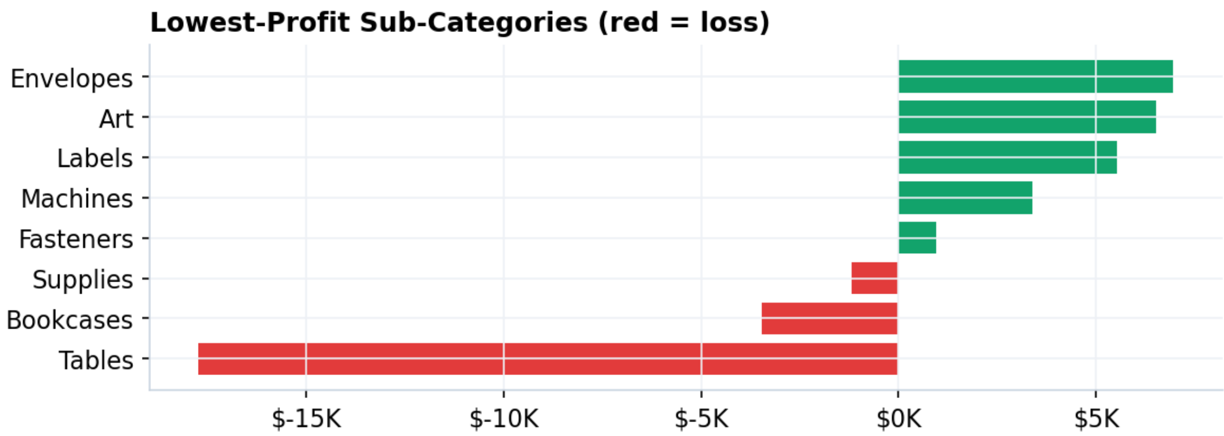
1. **Profitability — not sales volume — is the problem.**
Total revenue is healthy at $2.30M but margin is only 12.5%, below a 15% target. The business is selling well and keeping too little.

2. **Furniture is the margin drag.**
Furniture earns just 2.5% margin versus 17% for both Technology and Office Supplies. Three sub-categories — Tables (-8.6%), Bookcases (-3.0%) and Supplies (-2.5%) — sell at an outright loss.

3. **Discounting destroys Furniture profit.**
Loss-making sub-categories carry the heaviest discounts (Tables average 26%, Machines 31%, Binders 37%). Deep discounts push these lines below break-even.

4. **Central region underperforms.**
Central has the lowest margin (7.9%) and its Furniture line is negative overall, dragging down an otherwise strong national result led by West (14.9%) and East (13.5%).

5. **Revenue is concentrated.**
A small set of Technology products (copiers, binding machines) and top customers drive a large share of revenue — an opening for targeted retention and bundling.



Recommendations

ACTION	DETAIL	LEVER
Cap discounts on loss-making Furniture	Set a hard discount ceiling (≈15%) on Tables and Bookcases, especially in Central. Re-pricing these to break-even alone recovers an estimated $20K+ in annual profit.	**Profit**
Fix the Central region playbook	Audit Central's Furniture pricing and shipping costs; replicate the West region's lower-discount, higher-margin approach.	**Margin**
Double down on Technology & Copiers	Reallocate marketing spend toward Technology and Copiers (37% margin). Build bundles around top sellers like the Canon imageCLASS copier.	**Growth**
Plan inventory & campaigns around Q4	Demand peaks every November–December. Pre-stock high-margin lines and time promotions to capture seasonal lift without over-discounting.	**Ops**
Launch a top-customer retention program	Protect the concentrated revenue base with loyalty perks and account management for the top spenders.	**Retention**

Deliverables: cleaned dataset (CSV) · Jupyter notebook (.ipynb) · SQL query pack (PostgreSQL) · interactive HTML dashboard · this executive summary. Prepared as Portfolio 1 — Executive Sales Dashboard.